

Blueprint Sliding Door with Sequencer

OVERVIEW & PURPOSE

In this lesson you will learn how to create a set of proximity-triggered sliding doors as a blueprint, rather than using level blueprints.

EDUCATION STANDARDS

1. Understanding of how to use Unreal Engine 5
2. Understanding of basic Blueprint construction

OBJECTIVES

1. Door that slides open when the player approaches, and slides closed when the player moves away.
2. Creating a blueprint class that can be used in multiple situations.

MATERIALS NEEDED

1. Unreal Engine 5.2 or Higher

VERIFICATION

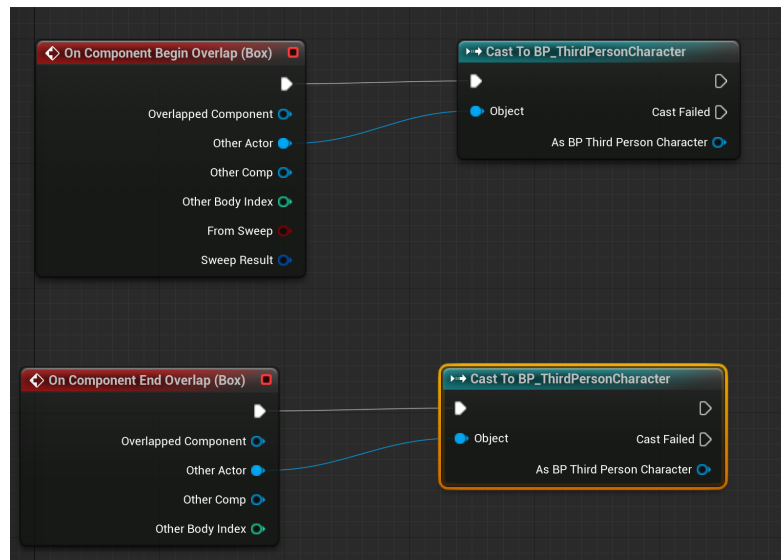
Steps to check for student understanding

1. Create another sliding door in another location, and show they both work

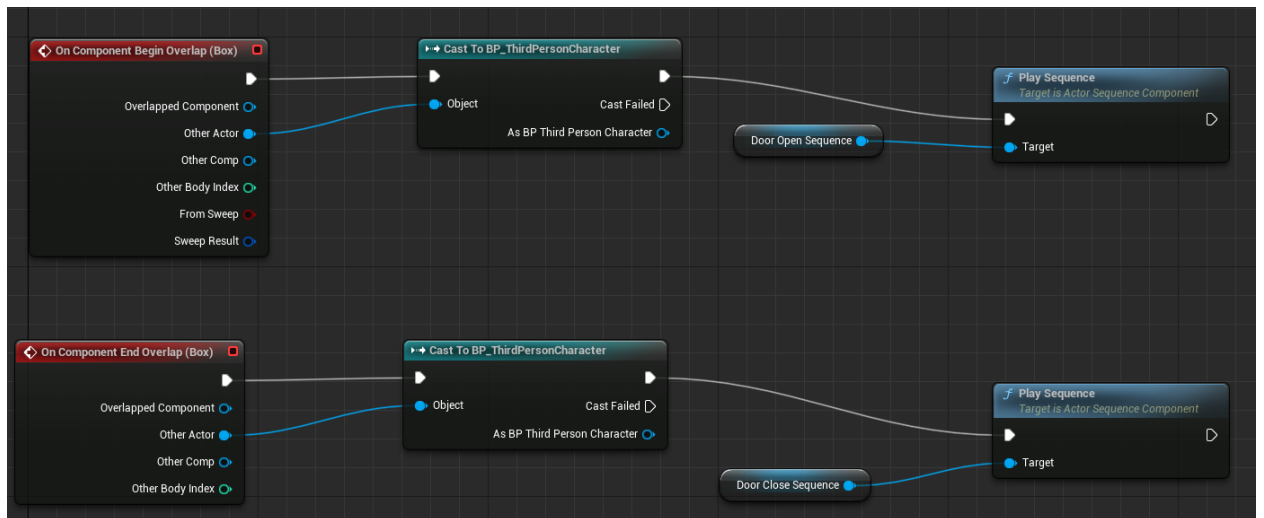
ACTIVITY

Describe activity that will reinforce the lesson

1. Create a new project with a Third Person template, using Starter Content. Name the project BlueprintDoorProject.
2. Create a hut using architecture doors, walls and pyramid shape from Starter Content.
3. Open folder ThirdPerson/Maps, right-click and create new Blueprint Class.
4. Choose class type "Actor".
5. Save and name the class BP_Door.
6. Open class.
7. Add the Static mesh SM_DoorFrame by dragging from the content area (under StarterContent/Props) to the DefaultSceneRoot. Rename this DoorFrame.
8. Add the static mesh SM_GlassWindow, and size it so it takes up the right half of the door frame. Name the component DoorRight.
9. Make sure the Mobility is set to "Moveable".
10. Duplicate DoorRight and name it DoorLeft. Position it appropriately.
11. Select DefaultSceneRoot and add a new component Box Collision. Resize and reposition the box so it covers the door frame and extends a little way in front and back.
12. Save the blueprint, then drag it from the content drawer to the scene, see that it looks correct.
13. Return to the blueprint editor. Select the collider box, and scroll down to the bottom of the Details bar. Click on "+" for On Component Begin Overlap and On Component End Overlap.
14. These are now added to the Event Graph. If there is anything else in the Event Graph, you can remove it.
15. From On Component Begin Overlap (Box), drag out a line and add a node Cast to BP Third Person Character. Connect the Other Actor line to the Object line. Do the same for On Component End Overlap (Box). After completion, should look like this:



16. In the Components tab, click add and search for "Sequence". Add the component "Actor Sequence" and name it DoorOpenSequence.
17. Select the Door Open Sequence and select "Open in Tab".
18. Click on "+ Track" and select Component -> DoorRight. This will add a track for the right-hand door.
19. Click on the "+" beside DoorRight and select "Transform" track. Open up Transform and Location to reveal X,Y,Z values
20. Make sure the red arrow is at 0.00, and click the + sign beside the Y to create a keyframe for that Y value.
21. Move the red arrow to the 5.00 mark.
22. In the viewport, slide the right door until it is hidden behind the frame (s.b. 50.0 units)
23. In the sequencer, with the red arrow at the 5.00 mark, click + again to register this new position. It should be -50.00.
24. Repeat this for the Left Door, which should be starting at 50 and going to 100.
25. Now make another Actor sequence named DoorCloseSequence. Repeat the steps so the windows now slide closed. (There is no reverse capability for actor sequences.)
26. In the On Component Start Overlay, right click in the area beside the cast and type in "Play Sequence (Door Open Sequence)" (or enough of the text so you can find the right node). Beside the On Component End Overlay, do the same for the Play Sequence (Door Close Sequence). Connect the exec tabs properly. After you are done, the event graph should look like this:



27. Now you can adjust the BP_Door sizing to fit the doorway (y=1.2).

28. Test out the scene. The door opens and closes properly!